第九章 虚存管理

1. A certain computer provides its users with a virtual-memory space of 2^{32} bytes. The computer has 2^{18} bytes of physical memory. The virtual memory is implemented by paging, and the page size is 4096 bytes. A user process generates the virtual address 11123456. Explain how the system establishes the corresponding physical location. Distinguish between software and hardware operations.

Given: VA: 32-bit PA: 18-bit Page Size: 4 KB = 2^{12} B ⇒ Offset: 12 bits

⇒ VPN: 20 bits PPN: 6 bits

Hardware: ① CPU: VA = 0x11123456 ⟹ ② MMU: VPN: 0x11123, offset: 0x456

③ MMU: Use PTBR to locate the start of PT in physical memory.

④ MMU: Memory access ⇒ Get the PTE ⇒ Get 6-bit PPN

⑤ PA: (PPN << 12) | Offset

Software: ① Page fault happens ② Loads a page from the disk to a frame, updates PTE

③ return and restart

2. A page-replacement algorithm should minimize the number of page faults. We can do this minimization by distributing heavily used pages evenly over all of memory, rather than having them compete for a small number of page frames. We can associate with each page frame a counter of the number of pages that are associated with that frame. Then, to replace a page, we search for the page frame with the smallest counter.

a. Define a page-replacement algorithm using this basic idea. Specifically address the problems of (1) what the initial value of the counters is, (2) when counters are increased, (3) when counters are decreased, and (4) how the page to be replaced is selected.

Define: Page frame → f, Counters → C

(1) When system starts, each C[f] is set to 0

When loads a new page, set C[f] = 1

(2) C[f] ← C[f] + 1

(3) C[f] ← max(0, C[f] - 1)

(4) ① Select the free frame.

② Otherwise choose the frame with the smallest counter,

③ If there is a tie, use FIFO

④ Replace it, and set C[f] = 1.

b. How many page faults occur for your algorithm for the following reference string, for four page frames?
1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8, 9, 7, 8, 9, 5, 4, 5, 4, 2.

Frame	f	f	f	f	f	hit	hit	f	f	f	f	hit	hit
A	1(1)	1(1)	1(1)	1(1)	5(1)	5(1)	5(1)	5(0)	6(1)	6(0)	6(0)	6(0)	6(0)
B		2(1)	2(1)	2(1)	2(0)	2(0)	2(0)	1(1)	1(0)	1(0)	1(0)	1(0)	1(0)
C			3(1)	3(1)	3(0)	3(1)	3(1)	3(0)	3(0)	7(0)	7(0)	7(1)	7(1)
D				4(1)	4(0)	4(0)	4(1)	4(0)	4(0)	4(0)	8(1)	8(1)	8(2)

	f	hit	hit	hit	f	f	hit	hit	f
A	6(0)	6(0)	6(0)	6(0)	5(1)	5(0)	5(1)	5(1)	5(0)
B	9(1)	9(1)	9(1)	9(2)	9(1)	9(0)	9(0)	9(0)	2(1)
C	7(1)	7(2)	7(2)	7(2)	7(1)	4(1)	4(1)	4(2)	4(1)
D	8(1)	8(1)	8(2)	8(2)	8(1)	8(0)	8(0)	8(0)	8(0)

Page faults = 9

c. What is the minimum number of page faults for an optimal page-replacement strategy for the reference string in part b with four page frames?

Page faults = 11 (I don't feel like to show the process)